Reflections on (Schumpeterian) Leadership:

A Report on a Seminar on Leadership and Management Education

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In this forum article we publish the proceedings of a seminar held at University of California, Berkeley, on November 17, 2003. The seminar brought together four distinguished scholars and business leaders to converse about leadership, not only from the perspectives of the different experiences the participants have in business, foundation, and academic leadership, but also with respect to substantive issues such as: What is laudable and what is despicable leadership? What is the role of leadership in business school research and curriculum? What is the relation between leaders and managers? How does governance temper leadership? The issues chosen are all central to the study and practice of leadership.

The panel began with the showing of the film, "Passion and Discipline: Don Quixote's Lessons for Leadership," produced by James March and Steven Schecter. March's film builds upon his ideas on leadership and organization (from a class he taught for many years at Stanford's Graduate School of Business) and demonstrates some fundamental issues and dilemmas in modern management/leadership theory. It portrays an intellectually stimulating interpretation of the contemporary meaning of Don Quixote by using interviews with modern leaders, supplemented by the portrayal and analysis of episodes drawn from the novel to demonstrate the significance of "joy, commitment, imagination, and foolishness." A main theme of the movie (and a main message for leaders) is the importance of sustaining creative, innovative, and sometimes foolish ideas and actions, especially in a world that rewards rational and conventional thinking and action. Leaders must sustain commitment to an idea, once

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envisioned, even when knowing that most creative ideas fail to produce successful outcomes.

Following the film, there was a panel discussion involving: John Reed, known for his long-term leadership role as CEO of CitiCorp (and most recently as interim chairman and CEO of NYSE); Eric Wanner, President of the Russell Sage Foundation; Professor James March, Stanford University; and Professor David Teece, UC Berkeley.

The following report of the proceedings of the seminar was prepared by Mie Augier and David Teece, and approved by the participants. Before the summary and transcript of the discussion, we will introduce briefly some of the issues discussed in the panel, some internal tensions in the field, and the relationship between leadership theory and practice. Finally, issues relating to the fundamental role of entrepreneurship in management/leadership will be discussed.

Leadership as an Interdisciplinary Enterprise

> "If the going is tough and the pressure is on; if reserves of strength have been drained and the summit is still not in sight; then the quality to seek in a person is neither great strength nor quickness of hand, but rather a resolute mind firmly set on its purpose that refuses to let its body slacken or rest."—Sir Edmund Hillary

> "On Tenacity: If you tell me why the fen appears impassable, I then will tell you why I think that I can get across it if I try."—Marianne Moore

Leadership is a compelling, yet elusive topic. It is one of the most important topics in management education and is also fundamental for the *practice* of management. Leaders play a critical role in building business organizations and in leading and organization to creating and capturing value from its capabilities.

One can trace the topic of leadership back to the thinking of Plato, Heraclitus, Odysseys, Aristotle, and Machiavelli. In many early treatments, leadership was about gaining and maintaining power. For instance, in the famous exchange between the philosopher Machiavelli and the Prince about leadership, Machiavelli advised: "Men ought either to be caressed or destroyed since they will seek revenge for minor hurts but will not be able to revenge major ones. . . . since love and fear can hardly co-exist, if we must choose between them, it is far safer to be feared than loved."[1] Such early views emphasized the *power* of the leader and did not require any compatibility of goals with the followers; a view that was challenged with Barnard's emphasis on the modern organization as a system of purposeful cooperation that requires a system of communication and some goal alignment to function.[2] Most organizations are systems of both conflict and cooperation; and leaders must be able to manage both.

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General acceptance of the idea that leadership matters invites an obvious question: What, then, is (good) leadership? In particular, the word "leadership" has some kind of burden of respectability when applied to understanding businesses, and perhaps for that reason it has accumulated a number of interrelated (and often inconsistent) meanings. Chester Barnard noted early that research on "leadership has been the subject of an extraordinary amount of dogmatically stated nonsense."[3] Likewise, Stogdill remarks: "Four decades of research on leadership have produced a bewildering mass of findings . . . It is difficult to know what, if anything, has been convincingly demonstrated by replicated research. The endless accumulation of empirical data has not produced an integrated understanding of leadership."[4] While the amount of "research" on leadership has exploded, we are nowhere near a "theory" or even a coherent understanding of the phenomenon, in part because the topic lacks definitional clarity and consensus with regard to fundamental understandings of "what is a leader" and "what is leadership."[5] Rost found 190 different definitions of leadership—surveying 587 books, chapters, and articles on the topic—most of which didn't do much in terms of distinguishing leadership from other important functions—such as that of the manager, the entrepreneur, or the administrator.[6]

Some well-known management scholars have provided important insights into leadership. Peter Drucker has stated that "the only definition of a leader is someone who has followers." However, he also notices its significance and, tongue in check, has remarked that "leadership is of utmost importance. Indeed, there is no substitute for it."[7] Philip Selznick, another original thinker in this area, talks about leadership as an organizational mechanism for achieving efficiency and control and for building commitment, understanding, and determination.[8]

In addition to management scholars, the phenomenon of leadership has attracted some attention from disciplinary perspectives by sociologists (Max Weber, of course, is the famous inventor of "charismatic leadership") and political scientists. While economists have traditionally focused on formal or contractual relationships within organizations, Hermalin has recently claimed that leadership can be accommodated within the rational choice framework of neoclassical economics through "leading by example," a mechanism by which leaders convince followers that they are not misleading them.[9] While there is some merit to this approach, it ignores the imaginative, innovative, and motivational dimensions of leadership, all of which are all crucial. Mark Casson has likewise connected leadership to business performance by noting the distinctive way leadership can work against agency problems.[10] Both Casson and Hermalin appear to reduce the function of leadership down to solving incentive-based principal-agent/transaction-cost type problems. While insightful, these approaches are inadequate. Leadership, as most other interesting phenomena in management research, is an interdisciplinary phenomenon, not readily understood using just agency theory or contractual theory.

The Webster dictionary defines leadership as "leading others along a way, guiding." This points to the importance of vision, of breaking boundaries and

setting new goals. John Gardner has hinted at the social aspects of leadership in his definition: "Leadership is the process of persuasion and example by which an individual (or leadership team) induces a group to take action that is in accord with the leader's purpose, or the shared purposes of all."[11]

As some (but not all) of the dictionary definitions indicate, it is important to understand leadership as much more than just the ability to lead or the activity of leading. Leadership it is both a dynamic competence and a relational process, which require specific skills and capabilities for envisioning, for setting aspirations, and for motivating others. Leadership embodies and elaborates a critical part of what modern strategy scholars call "dynamic capabilities," which, very briefly, refers to the firm's ability to sense and to seize opportunities and threats, and to exploit internal and external firm-specific competences.[12] The leader must manage and direct this function, using his or her personal vision, skills, and reputation to bring others along. It is an organizational competence, but it must be directed by a compelling personality. Effective leadership draws on many individual and organizational skills. It is particularly crucial when there is significant uncertainty and many possible paths forward. The leader will help define alternatives, select among them, and execute them. There's an important element of imagination here, which is consistent with Joseph Schumpeter's emphasis on vision as necessarily preceding (entrepreneurial) action and leadership: "We should first have to visualize a distinct set of coherent phenomena a worth-while object of our analytic efforts. In other words, analytic effort is of necessity preceded by a preanalytic cognitive act that supplies the raw material for the analytic effort. . . . this preanalytic cognitive act will be called Vision."[13] Seeing leadership as a dynamic competence stresses the importance of vision to first imagine an idea or a project; commitment to execute it and thus bringing imagination to work; and pleasure and joy in the process of building better competencies—dimensions that are discussed by Professor March and the panel below.

Schumpeter's views of the entrepreneur nicely encapsulate our idea of leadership as an inherently entrepreneurial activity. The leader/entrepreneur is motivated by the "dream and the will to found a private kingdom"; the "will to conquer: the impulse to fight, to prove oneself superior to others"; and the "joy of creating, of getting things done."[14]

Because leadership is critical to business enterprises, particularly those operating in highly uncertain environments, the study of leadership is central to the mission of a great business school; and business schools must help shape individuals into understanding both the theories and the practical implications of leadership. This is not an easy task, in part because leadership is often ill-defined as an academic and practical subject. Seeing and analyzing leadership as a dynamic competency and emphasizing its evolutionary (Schumpeterian) elements can help advancing the treatment of the topic.

Furthermore, none of the existing contributions to understanding leadership examine (or provides a framework for examining) the central questions of leadership, such as, what makes good leaders? Or, what's the difference between

a leader, an entrepreneur, and a manager? Moreover, the field of leadership, like other fields of management education, confronts the following more foundational problems/paradoxes:

Being Interdisciplinary, but Being Disciplined

A fundamental requirement for management education is to maintain a balance between interdisciplinary research and drawing from disciplines. Business schools are barely a century old, and recognizable research within them dates from the early 1960s. One of the first journals, *Management Science*, was more game theory than management; and subsequent journals have added to the confusion of the different possible approaches to business school research.[15] Business practice presents real problems and issues; yet a general discipline of business with a recognized set of axioms and predictions has not so far emerged. Perhaps it never will. Hence, scientific advance and scientific insights must be drawn—if they are to be drawn at all—from neighboring disciplines, such as economics, psychology, and sociology. Inasmuch as business and management issues transcend disciplines, so must scholarships in business schools transcend disciplinary boundaries.[16]

Accordingly, a fundamental requirement for management education is to maintain a balance between disciplinary and interdisciplinary research. To provide insight into the complex problems that businesses face, we need to be interdisciplinary because the problems themselves are interdisciplinary.[17] However, sometimes interdisciplinary work becomes multidisciplinary; and multidisciplinary work in turn may lack a disciplinary foundation altogether. This can lead to pluralism and fragmentation.[18] Gurus emerge pushing one or another popular remedy to perceived organizational problems. Such fads and fashions, often fueled by management consultants, do not lead to scientific progress. This is a problem that leadership (as well as other sub-areas in business education) faces, too.

Being Scholarly, but Being Relevant

Management research needs to be grounded in practical problems, but we still need good basic research. There is a lot of pressure to produce consultant-like work, dressed up as light articles, and to be "relevant." We need basic research that can help us to think through and frame—*not* necessarily solve—management problems. Framing fundamental issues and finding good but "messy" research problems is often more valuable than tight solutions to narrow, well-defined problems.

The quest for fundamental knowledge on the one hand and the search for relevance on the other have animated business education for the last 50 years. As both scholars and leaders have noted,[19] interaction between the academic community and management practice is important—but perhaps scholars in management theory should be less concerned about achieving immediate relevance and more concerned about providing a basic framework for understanding managerial problems. Jim March has long argued that "research is a

testament."[20] In the same spirit, John Reed, has provided the following insight:

> In general, I believe it is very important for business practice to be shaped by basic research and basic knowledge. Research that draws from the academic disciplines (but in an interdisciplinary way) serves the important function of creating a framework that allows managers and management practitioners to understand the context and content of the specific problems they are dealing with. So, clearly, the interaction between the academic community and management practice is important—and management theory need to not worry so much about immediate relevance as to providing the basic framework for understanding managerial problems.[21]

In other words, academic researchers should serve as those who disseminate and apply knowledge in the pursuit of better theories and of better understanding of how leaders, organizations, and business practices work.[22] Leaders and business practitioners should embrace the resulting theories while continuing to explore new business models, inventing new leadership tools and organizational forms that scholars can then use to guide research, pushing science forward.

The Evolution of Management Education in the U.S.[23]

When business schools began in America over a century ago, it was initially Wharton and the University of California at Berkeley that led the way. Early business schools developed their roles to provide managerial training for railroad executives—the first big businesses to emerge. In America, early business schools had difficulty establishing academic legitimacy. Fortunately, however, several scholars did start on university campuses, providing a future opportunity to reach out to scholars in other disciplines.

However, early business schools in America were not considered as serious participants in the world of academic scholarship and intellectual pursuits. They often defined their role primarily in terms of codifying and communicating good business practice, as exemplified by business case writing and teaching. Indeed, many early faculty positions were filled with experienced businessmen (e.g., Arthur Andersen) rather than young scholars. Despite the efforts of some deans and faculty to migrate towards more academic pursuits, the schools emphasized practical, not theoretical courses; applied, not basic science; and the contributions of faculty were more often publishing in practitioner magazines than in academic journals. Herbert Simon, who witnessed and contributed to the transformation of business education, reflected in his autobiography: "Accurately or not, we perceived American business education at that time as a wasteland of vocationalism that needed to be transformed into science-based professionalism, as medicine and engineering had been transformed a generation or two earlier."[24]

The immediate post-war period was an era that glorified big science.[25] The social and behavioral sciences became more quantitative, more analytical, and more committed to scientific principles. A report by Robert Gordon at Berkeley

and James Howell at Stanford defined a watershed in business education when it advocated the adoption of analytical approaches to management education.[26] Spurred by the Gordon-Howell Report, the Ford Foundation dedicated more than 35 million dollars during the 1960s to successful efforts to reform business schools.[27]

The Ford Foundation had at that time formulated a program for "the study of man" (which became known as "the behavioral science research area") the specific objective of which was stated as: "The Ford Foundation will support scientific activities designed to increase knowledge of factors which influence or determine human conduct, and to extend such knowledge for the maximum benefit of individuals and of society."[28] Research had to be scientific; embodied in the Ford Foundation's understanding of the behavioral science concept was "its emphasis upon the scientific approach to problem solution."[29] It had to be practical, to some extend at least, given the foundation's interest not in knowledge per se, but in "knowledge which promises at some point to serve human needs." Furthermore, it explicitly encouraged interdisciplinary research. "The program is interdisciplinary and inter-field. Its goal is to acquire and apply knowledge of human behavior, and segments of all fields and disciplines will make contributions in varying degrees."

Carnegie Mellon University (then, the Carnegie Institute of Technology) through its Graduate School of Industrial Administration became the role model for a research-based, disciplinary-oriented (but very interdisciplinary) approach to business education and an invigoration of fundamental interdisciplinary research in accounting, finance, marketing, operations research, microeconomics, and organizations. March and Simon's *Organizations* (1958) and Cyert and March's *A Behavioral Theory of the Firm* (1963) are two significant results of the early work on business research at Carnegie.[30] In addition to filling a need in the establishment of the behavioral sciences, research on organizations became the emergent discipline of business school education, bringing together different disciplines in the study of decision making and behavior in organizations.

In the early years of this new epoch, the interaction between disciplinary specialization and interdisciplinary collaboration was essential to the development of management theory. Management theory at Carnegie was interdisciplinary, but in a disciplined way. Scholars studied business phenomena from the perspective of the established social science disciplines. Now, while this provided opportunity for research in management education in an interdisciplinary way, it also came at some cost. Being a stepchild of the pure behavioral sciences, early business education could borrow from them all, yet belonged to none, and it didn't immediately have a home as a recognized science itself. It was very clear early on that a general theory was not in the offing.

The 25 years that followed the Gordon-Howell Report saw tremendous progress in business schools, and the emergence of scholarly traditions. However, what was conceived by the Carnegie Scholars as an interdisciplinary challenge devolved into the domain of monodisciplinary scholars—with neoclassical economics in particular coming to deeply infuse business school curriculum.

The situation was such that by 1980, at some U.S. business schools one could frequently not tell the identity of a scholar in a business school from a scholar in a discipline-based academic department (such as economics, operations research, or sociology). The pendulum had perhaps swung too far, and relevance had been lost—at least in some academic groups in some business schools. The consequences were unfortunate. In the U.S., there were serious problems with financial institutions (the Savings and Loan crisis in the 1980s) yet business schools couldn't field relevant faculty or offer relevant curriculum, as knowledge of financial institutions and past financial regulation was extremely limited. The Savings and Loan crisis was due to poor regulatory design and poor management. Likewise, American manufacturing was in competitive decline, yet business faculty couldn't explain the wide dispersion in the performance of firms in the same industry using the same technology.

Indeed, since the mid-1980s, U.S. business schools have attracted and nurtured faculty who are more applied in their focus—although many are reluctant to grapple with management issues and (at least the economists) tend to stay in more comfortable and familiar areas such as public policy. Nevertheless, tremendous progress has been made in making business schools more analytical and methodologically rigorous. As a result, the quality of business school research is today markedly better than it was 3 decades ago.

Towards an Emergent Discipline of Leadership?

James March has discussed leadership issues on several occasions, and his film "Passion and Discipline: Don Quixote's Lessons for Leadership"[31] is an extension of his examination of the problems of achieving a balance between "exploitation" and "exploration."[32] Exploiting existing capabilities provides rewards in the short run, but it doesn't prepare people for the inevitable changes in technologies, capabilities, desires, tastes, and identities. For such preparation, exploration is necessary. Exploration involves searching for things that might come to be known, experimenting with doing things that are not warranted by experience or expectations. In research as well as in life, we need both sides.[33]

In March's words: "We live in a consequentialist world. We are always justifying things by their consequences. But many of the more important human actions are significant precisely because they are not justified by their consequences."[34] March uses Don Quixote to argue that sometimes, the essence of life cannot be understood in conventional (consequentialist) terms. The essence of action requires more than reason—it requires foolishness, imagination, commitment, and unjustifiable faith. March quotes Quixote: "For a knight errant to make himself crazy for a reason warrants neither credit nor thanks; the point is to act foolishly without justification." In this perspective, action becomes truly valuable (and human) when it is not justified by consequences.

On a theoretical level, it is too early to talk about an emergent paradigm of leadership. The fragmentation of the fields in business schools has resulted in several barriers to the development of more integrative research on leadership.[35]

To add further complexity, stimulating the generation of new ideas is not easy, because it involves increasing the generation of bad new ideas as well.[36] Alfred Whitehead once noted that "great advances in thought are often the result of fortunate errors."[37] We don't know in advance which ideas will turn out to be great, and not just crazy—but we must persist in stimulating and supporting the imagination and creativity underlying both innovation and (great) leadership. Nurturing imagination and innovation, however, requires organizations and individuals to be tolerant to ideas that are deviant from the norm and may, at first, seem crazy.

Although Plato established a school for leadership (the Paidea) in early Greece, most of our current understanding of leaders comes not from philosophers, but from an examination of political and military leaders: Henry V's victory at Agincourt against overwhelming odds; George Washington who defeated better-equipped Hessian forces; and Winston Churchill who kept up British spirits and capabilities during the Battle of Britain. But Hitler and Mussolini were leaders too. The complexities and possible sinister side of leadership highlights the importance of values, personal integrity, and systems of institutional governance. To assure such mechanisms, leaders in business organizations at all levels and functions must view leadership as part of the company's capability. They must also, as emphasized by March, have a strong "sense of self."

There is an important difference between managers and leaders. Managers seek order and control, leaders tolerate some degree of chaos and lack of structure. This is related (but not identical) to the distinction made by Max Weber between leadership and authority.[38] Entrepreneurs are more likely to be leaders. Steve Jobs, for example, has demonstrated "joy, commitment, imagination, and foolishness" at Apple. Entrepreneurs frequently require managers to help them realize their dreams—raising the question of whether leadership might reside in a team (or the organization) rather than in an individual. As Schumpeter noted: "The entrepreneurial function may be and often is filled cooperatively—in many cases, therefore, it is difficult or even impossible to name an individual that acts as 'the entrepreneur.'"[39] These visions and others pointed out by the panel below are significant to the future of leadership as an academic and a practical topic.

A Panel Discussion on Leadership

John Reed: If you step back and think about leadership in conventional terms, I would start out simply by saying that we *can* learn about leadership. People say they can teach leadership; I think probably the military can do it and can do it well. It's not clear to me, however, that we teach leadership in business schools; nor do we do it when you join a company and start a business career. But we can observe leadership and we can make discuss it. Jim [March] did so drawing on literature.

My own sense is that as you get down to the basics, leadership is about *capturing attention* and *motivating* people to follow your way—your vision and

your dreams. You can capture attention because you're in a position where people listen, but sometimes you have to capture attention in other ways.

I always ask people who are talking about leadership: "Imagine that tomorrow you were put in charge of the Chicago school system. How do you capture the attention of the people who are busy, trying to run the school, trying to teach the students? You are the new superintendent—how do you capture everyone's attention? And, if you can capture their attention, how do you motivate them to pay attention."

It could be through inspiration. Martin Luther King, Jr., had the gift of imagination that he could verbalize. That motivated our society—not just us as followers, but our whole society—to try to convert the dream into reality. But as a leader this issue of capturing attention and motivating performance is central.

Jim's contribution by reaching back to Don Quixote and talking about "knowing who you are"—and, most importantly, not only knowing, but *behaving* in a way that is *consistent* with who you are—is very important for us as leaders (and as human beings). There are many who know who they are but they enter into a situation which they call "work" and they try to check who they are at the door. They try to assume a role that they perceive to be appropriate in the situation. But it doesn't in fact reflect who they are. And over a period of time, my guess is that their ability to persist in a leadership role that is fundamentally artificial gets pulled away.

If you're going to be in a leadership role you have to reach into yourself for some form of self-assurance that gives you the ability to take on difficult responsibilities. I think that if you look around in the business world, you'll see a lot of people with big egos. You often wonder how could people with egos of that sort be in a position of responsibility?

Frankly, my observation, having witnessed this over the years, is that sometimes the egos are necessary for the individual—to give him or her the self-confidence to do what they are doing. And they need that. They rely on that false sense of themselves that we call an ego in order to be able to persist in playing their role and meeting their responsibilities.

Others have a sense that what they are doing is right and that they have the duty to do it. I suspect Martin Luther King Jr., and Gandhi weren't drawing on their egos, they were drawing on a sense that came from within—on a sense that the struggle they were leading was something that was "right" in some deep form. Obviously, if you can draw on deep moral strength, as they did, it gives you tremendous self-assurance. Many leadership situations do not quite end up as intended, and we must persevere and be patient with our ideas. If you can draw on a deep moral sense, it clearly helps.

Some of us draw on a sense that is fundamentally *analytic*—you go to a business school and you learn how to make rational decisions and you work in business for a number of years and you get to know the details of the business and you draw on what you consider to be the deep knowledge of the situation. That gives you the self-assurance to persist in playing the leadership role. But

it's a sense of analytical strength that you have and knowledge of what you are doing that tends to run out if you moved into a field that is not directly related to that for which you were trained or in which you've had experience. If the world changes around you, and you find you have to deal with discontinuities, your leadership skills can be severely tested, because the source of your self-assurance is no longer there. You begin to fall back to some of Don Quixote's "irrational" beliefs in himself in order to persist in the leadership role.

Many draw their own sense of leadership from the imperative of the goal. You know, "we need to earn 12 dollars and fifty cents a share," and we come to believe that you draw your sense of yourself as a leader from your commitment to that goal. "We're going to have 50 billions of dollars in sales" or "we're going to have a market share of 15.5 %." The problem with that kind of thinking is obvious: Good leadership is not grounded in anything other than the goal. You don't have a lot of resources to draw on if the only thing that is the basis of your assurance as a leader is your determination to get to that goal. Determination helps; and having clear goals clearly makes you feel good about what you're trying to communicate. But if there isn't anything beneath it that touches something more fundamental, my experience as a leader is that, without those values you can find yourself short and then you have little to draw on.

Eric Wanner: I've spent a lot of the last 20 years of my life in science funding and, certainly, its supposed to be a very serious business. We would probably never admit there was anything "Quixotic" about it. We are using either public money or private accumulations in order to fund certain lines of scientific activity and not others, and this is driven by peer review and consensual processes having to do with the merit of work. It's supposed to be very consequentialist—and very responsible. But if you try to think about how it applies to what you do, you really have to admit that (more than you would want to reveal) it is actually quite "Quixotic." I'll try to give you some examples.

The issue of *vision* is obviously paramount. Science goes on doing what it does, and the question is whether or not it needs or deserves any exogenous shocks that destabilize it in any way. Of course, some of these shocks can come from plain genius; there's nothing we can do to promote it or even explain it. But others can come from some kind of inspired view from outside of what science is up to and what it might be, but isn't yet.

For example, I think of a man named Warren Weaver whom some of you may know. He was quite a mathematician in his own right and a program officer at the Rockefeller Foundation in the 1940s. Warren Weaver had an idea, which is that the tools of organic chemistry could be used to analyze nucleic acids—these big molecules that we eventually all understand contain knowledge about the big secrets of life. He went to Linus Pauling and said, in effect, "Hey, stop doing organic chemistry. We don't need more of that. We need you to look at these nucleic acids." And in the course of his work during the 1930s, Weaver actually funded something like 15 Nobel Laureates before they got their Nobel Prize. And in some ways he had the arrogance and the vision to turn science in a direction that it might not otherwise have gone without two things, first, his

vision, and second, the fact that he commanded certain resources. That to me is the kind of story that I take home with myself at night and say, "Dare to be a Warren Weaver."

It is, of course, a hard thing to do because you run into all the sorts of very intelligent criticism that science will bring to bear on any effort to destabilize it from the outside. So science is itself a collective effort to critically evaluate ideas and to find out what's wrong with them. If you follow a Popperian view of science, you'll know that what we're trying to do is to let all the flowers bloom but then throw away all the bad flowers and leave just the good ones, or the ones that we haven't yet proved are bad. If you have a new idea, you want to start something, the way Warren Weaver did—then you have to defend it before it is defensible. You have to say "Hey, science. Look in that direction, go that way, explore that unexplored territory."

You have to have sufficient enthusiasm and sufficient ability to sell and market that prospect—and, of course, it helps to have some money to back it. In the course of doing that, you obviously have to be committed—not unlike Don Quixote. There's a period that I think of as the "sea-sick" period: you have an idea, and you convince some other people that it is a good idea. And you start funding some prospective work, scientific work. And then you begin to wonder in the dark of night, is this really such a good idea after all? You can't prove it to yourself or to anyone else. You're still waiting for the results of that early work to come back.

I once heard a wonderful comment about leadership from a man who was president of Russell Sage long before I was, David Truman. He defined one of the functions of leadership as *absorbing uncertainty*. That struck me as very, very true. There's a period in which there are all kinds of rational reasons to undermine an idea, to back off from an idea, and yet what a leader has to do at that point is to say, "Yes, these uncertainties are real. We may fail." But somehow you have to absorb those and make the case for going ahead.

Now, there's also something that is non-Quixotic about science funding. The thing that strikes me about Don Quixote is that he just didn't know when to stop. He keeps fighting all these battles on and on. At some point, at least in this world, you have to say "enough." In fact, in the market—in John [Reed]'s world—the market tells you when it is enough. If you are loosing money over a long enough period of time, perhaps you have to loose money for quite a while, there is a market signal there. In a political system, in a democracy, votes tell you where to stop. In my world, you have to look at the market place of ideas. Eventually, if something is working, as it was for Warren Weaver, the biologists took it away from him and they created their own science. That's what we are trying to do. But if they don't take it away from us, we had better stop.

So, these are some Quixotic and non-Quixotic parts of my world as I understand them. It is nothing as beautiful as Jim's world but I think it has some similarities.

I wanted to leave you with one provocation. I do think there is a big, unanswered question. In fact, Jim and I have had that conversation many times,

about whether when you are managing for innovation, you try to manage for a distribution of outcomes which is very tightly clustered around the best mean you can imagine, or whether you are willing to sacrifice some control over this process of generating innovations so that you will get a greater variation. And if you do, are you willing to sacrifice some downward decay in the mean in order to get this wider distribution of outcomes, which may have some real gold in the upper tail. We never know exactly how to play that and I think there are some innovators, some science managers, who go one way and some go the other. When Jim was the chair of the Russell Sage Foundation, there was a lovely tension because one of us went one way and one went the other. I leave it to you to figure which was which.

But, here's my provocation. And its because of Jim using all these wonderful literary references, I want to show you that I've read some books, too. There is another wonderful book about a servant and master, written about 150 years after Quixote by the French Encyclopedist Denis Diderot, called, "Jacques le Fataliste." Milan Kundera, the Czech novelist, wrote a drama version of Diderot's tale called "Jacques and his Master." Jacques and his Master go through many adventures, usually having to do with some outraged husband, and at the end of the play they are on the run again. The master is an aristocrat and totally hopeless when it comes to dealing with the real world, his servant Jacques understands about how to master fate—or at least escape. The master says to Jacques, "Look, we're in trouble. Which way are we going to go?" Jacques says, "Master, we should go forward." The master says, "Yes, but which way is forward"? And Jacques says, "Don't you understand, this is one of the great inventions of the modern world—all directions are forward."

DAVID TEECE: It is with some trepidation that an economist will take up the subject of leadership, but I must. I have reached the age of fifty, so I now feel qualified at least to address the issue but not necessarily to come up with the right answer. After all, as Jim explains, the critical issues of leadership are indistinguishable from the critical issues of life, and there's a lot about life and living that is embedded in leadership. And it is not something that yields easily to analytical frameworks or to two-by-two matrixes, for that matter.

Most of what we know about leadership comes from the study of political leaders and military leaders. It is only recently that we have started to focus on business leaders. Most of our lessons come from Mahatma Gandhi, George Washington, Winston Churchill, Ronald Reagan, and other great names in history. And that's why I find Jim's framework particularly compelling because it actually works—for understanding business leaders, military leaders, or political leaders.

Imagination.[40] Obviously, this is critical and it is not just for what leaders do, but also for what entrepreneurs do. In fact, *Seeing Things Differently* is the title of a book on entrepreneurship and innovation.[41] In the innovation process, you can look at things and see what you want to see and other people look at the same things and see something different. Being able to see different possibilities for a technology or being able to see a different business model is very much an

imaginative act. I think it's well-recognized that imaginative, out-of-the-box, creative thinking and persuading others of the merits of the new perspective is a critical component of entrepreneurial leadership.

Commitment. This is hard to teach in a classroom environment. What we *do* know is that in business formation, uncertainty is high and failure is the norm. What distinguishes leaders from others is their ability to constantly pick themselves up and continue on, and constantly be moving forward. Whenever one endeavors to build anything of significance, there will be setbacks. Leaders help work through these barriers and move forward.

Joy. There is the importance of *processes* and simply enjoying the journey. Given that the destination is so elusive and you won't necessarily get precisely where you want to go, let's hope that the journey is joyful. In this way, persistency can be sustained, and positive outcomes achieved.

It is useful to think about some of the differences between management and leadership. Some great managers are leaders—John Reed is one example. But many managers are not leaders. And, in fact, management science as we know it is not about leadership; so we need to keep the distinctions clear. Management is usually about control and it is usually about structure. It's not about chaos, and it's not about overcoming adversity. So, a "good" training in management may not get you where you need to get in terms of leadership. And if you look at the military, you don't manage an army in battle—you lead. And you lead from the front, not from behind. So there is a very important distinction there that I think needs to be spelled out and it ought to influence curriculum design and content in business schools.

Entrepreneurs have many things in common with leaders. Indeed, I believe that the entrepreneur has more in common with the leader than managers do because the entrepreneur takes joy in seeing cool, beautiful technology succeed in the hands of customers or in seeing new ways of business excite people's imagination and capabilities.

With this as background, let me mention a few additional elements that must enter the framework to convert leadership into a positive social force.

Governance. Hitler in Germany, Mussolini in Italy, Hussein in Iraq, and Fastow at Enron took their leadership abilities and harnessed them in ways that were corrupt, selfish, and in some cases evil. How do we prevent leaders from using their skills for immoral or improper purposes? Posing this question brings forward a possible necessary tension that must exist between governance and leadership. Leadership skills must be channeled and checked in the political context by the instruments of political governance (e.g., the President's power is limited by Congress and the Courts); and in the case of CEOs of business organizations, by the instruments of private governance (independent directors, auditors, and the legal and regulatory framework) and also by capital market competition, which also serves to check the behavior of management.

Tempering Imagination. It should be obvious that not every dream is worth pursuing. Eric [Wanner] gave some excellent perspectives on this. Some ideas

should be abandoned. Indeed, despite the reality of rapid technological change, most new ideas never develop into something important. There is a need to temper one's dreams with some degree of judgment and experience. In the business context, dreaming about something customers will never want is likely to be unproductive. Tempering creativity with just a modicum—but no more than a modicum—of reality is what separates a dreamer that leads from a dreamer that just dreams.

Foolishness. A subsidiary theme is the value of foolishness—and Don Quixote has his fair share of it. Here I believe it is important to distinguish between ex ante and ex post foolishness. If one is not willing to suffer jabs, insults, and mockery for ones ideas, they are probably not bold enough. But ex post they need to look brilliant, not foolish. Many apparently foolish ideas eventually turn out to be brilliant. For instance, the use of electricity for home lighting and heating was once considered to be unnecessary and unsafe.

Don Quixote offers many lessons for leaders. Another person is Ulysses, or Odysseus as the Greeks call him. Alfred Lord Tennyson, in a poem of the same name, describes the soul and the spirit of a seeker who yields at nothing. In the poem, Ulysses, like Don Quixote, has imagination and commitment: "Yet all experience is an arch wherethro'/Gleams that untravell'd world, whose margin fades/for ever and for ever when I move." Ulysses is "always roaming with a hungry heart" and while death closes all, "Some work of noble note, may yet be done."

So, I agree with Jim that examples from literature can enrich our understanding of leadership. Literature is a lens through which we can examine life.

Questions from the Floor

Audience Member: With all the science and literature that we have on leadership and management, why do we need to go to fiction?

James March: I can try to explain for myself, but others have to speak for themselves. As a scientific concept, leadership is a mess. It is not a good scientific concept. If you look at the scholarly literature on leadership, and if you are a serious scholar, you are likely to feel some combination of embarrassment and revulsion. There are many non-scholarly books written on leadership that are even worse. Why is that true? It is not because these are incompetent scholars and it is not because people are dumb. I think it is because we are confronted with an enormous myth that is extraordinarily important in our society, tied to a need to imagine that humans control their own destinies. It is a myth that is recorded as far as back as we go in oral history. We tell history in terms of leadership. It is a comforting myth and one that can be used to tell stories of history after it occurs, but it is not a myth that really gives us much in the way of predictive understanding. But as I came to teach leadership, I came to believe that the interest in leadership came less from an interest in understanding history than from an interest in understanding how to think about being a leader. Now, there are many kinds of understandings that we've come to over time and those are

embedded often in great art and great literature. It is primarily a testimony to my innocence that I tried to teach a course on organizational leadership at Stanford for many years. I came to the conclusion that the critical issues of leadership were indistinguishable from the critical issues of life, and that the critical issues of life are addressed best in great literature. So, for a number of years, I taught a course where the reading assignments were Shakespeare's "Othello," G.B. Shaw's "St. Joan," Tolstoy's *War and Peace*, and Cervantes' *Don Quixote*. I think that somewhat formidable reading list provides material for discussion of some serious issues about leadership.

If we ask, "What are the fundamental issues that we are trying to understand?" we discover that they are only partly scientific issues. If we ask the question, "What justifies great action?" which is a very fundamental question in leadership, that's partly a scientific question in the sense that we can ask what are the justifications that people use. But it is also a philosophical question, it is an ethical question, and it is a moral question. And those have to be put together. And they are put together very well, in my judgment, in great literature. So I started to ask, what are the issues that people really wanted to discuss in leadership? These were issues such as, "What is the relation between private life and public life?"—an issue that I think is well discussed in Othello. "What's the relation between genius and madness?" All of these questions have scientific understandings that one wants to weave into the discussion. But the first text is often drawn from literature. What is the role of leadership in general? If you want to talk about that, you should, if not start with Tolstoy, certainly involve Tolstoy.

In addition, there was a secondary proposition underlying my efforts and it is certainly not demonstrable. It is that our leaders—our political leaders, our business leaders, and so on—and we, as we contemplate leadership, should get into the habit of reading great literature. I said to my students, "You give me any piece of great literature" (and they quickly learned that meant something written more than 100 years ago), "and I'll show you how it is relevant to modern leadership." And we played that challenge and I won (some of my students may not think I won). And we did the same thing with poetry. Show me any great poet and I will show you his relevance to leadership. But the meta message is: these are the things that engage minds in an evocative way that pop-psychology or pop-leadership books simply don't. They don't give you the 30-second introduction on how to do something—they don't do that. But, they make you think, and that's not a bad idea.

Audience Member: You don't distinguish between power and leadership. Do you see certain leaders as having power but not being a leader? Are there different characteristics? Don Quixote was certainly not a powerful man.

James March: Quixote was not a leader at all and he is not a great model for leaders. He, however, teaches things that are relevant to leadership. I'm reluctant to comment on the power issues, as some of you may know, I've said for many years that power is a terrible concept. I've also said that leadership is a

terrible concept, so how can I put two terrible concepts together? [laughs] Power is the political scientist's label for his unexplained variance, basically. Power is to the political scientist what personality is to the psychologist—what you call everything you don't understand. Everybody has to have that. Economists call it "utility" [laughs] and it is very useful, but it doesn't really explain much. Some of my wiser friends in the panel may have a different answer to this one.

Audience Member: But when you look at leadership in history, do you see the great leaders as having great powers too?

James March: As I understand it, you want to give "leadership" a more positive loading than you give to power. So the question is, do I personally see some people in positions of power whom I admire more than others? I think I admire people who know who they are—you have a sense of that inner integrity. I also admire people who can think rationally. The Quixote point depends upon the fact that we live in a society which greatly exaggerates the merits of rational and consequentialist thinking. But that doesn't mean that consequentialist and rational thinking isn't relevant. I think it is very relevant. It is just that you have to balance it. So, I admire people who can do both. I've said fairly frequently that "leadership is a combination of poetry and plumbing." The leaders that I admire keep the toilets working and also express themselves and, even more, lead other people to know who they are and what to do.

Audience Member: This question is mainly for Jim, but I'd also like John to comment on it, if you would. My question is: Does leadership transcend morality? Is leadership amoral? We have recent examples of Bin Laden and Saddam Hussein whose leadership remains in place even though they aren't. And Mussolini. All these people were able to embody the Quixote-like characteristics and yet, they are going towards the darkness. So, is this leadership amoral?

James March: In many ways you touch on what David [Teece] talked about and that is if you take the notion of one who knows who he is, and that is important for a leader, then you've just unleashed—you have risked unleashing—a leader from any reality concerns. And that has to be balanced in my judgment and as David said, with some kind of governance system that says, "OK. You're committed, but you are wrong." And without that, you're going to have a lot of committed "knowing who they are" people who are headed in horrible directions. I think your notion of transcending morality is also interesting. Because one of the things that great leaders do is to challenge our morality, to say "I have a vision, or I have a notion of what is morality that is different and I'm going to try to reshape your notion of morality." That's what great religious leaders have done and that's what many great political leaders have done. So, it's a little like Eric's observation about scientists. You want those scientists who have nutty ideas, but you have to recognize that most nutty ideas are just nutty. In order to get the few good ones, you have to bear the bad ones, but you may need some kind of device so that bearing them doesn't mean total disaster.

JOHN REED: It is a complicated question. There clearly have been leaders in history who were fundamentally evil but were nonetheless gifted in their ability to create a vision and to get followers. Fortunately for the world, eventually the overwhelming evilness of it turns out to be the weakness and they tend to collapse—otherwise we might be living in a world that was led by such a person. Clearly, you can't separate out leadership as a capability from the morality underneath because there's so many examples of quite successful leaders who we would think have lost their morale compass. In the business arena, it is fortunately much more constrained. What we've experienced recently is a by-product of greed, ego, and loss of balance. And now society has decided, and I think correctly, at least as a first response, that appropriate governance would be an appropriate restraining mechanism for this kind of misbehavior. I don't think, however, that many of the instances about which we read were about particularly outstanding leaders. The business area is a much more mundane environment. And we certainly saw big egos. Leadership that is based on personalities tends, in my mind, to have some built in limitations. The person I think of as most exemplary is [Lyndon] Johnson, who was a great leader of the Congress and drew his leadership skill from his ability to manipulate people. I don't mean that only in a negative way, but to manipulate them to do good, as with the Civil Rights Act and the breakthrough from 150 years of southern resistance to discussion of the subject. But interestingly, when he ran into a more complex situation where his skills as a personal leader simply couldn't play—the Vietnam War—the limitations of that particular style of leadership became evident and resulted in a limitation. So, the direct answer to your question: Yes, you can be a great leader without a moral compass, unfortunately. We have tons of examples in the business world. Fortunately, the arena is somewhat more circumscribed and the damage is different. But I must say, one thing that I've learned is beware of companies that are run by personalities, because they have a tendency to overrun themselves. And this governance issue, of course, is an approach to trying to restrain that.

AUDIENCE MEMBER: Where in leadership do you see integrity?

ERIC WANNER: It seems to me that you have to distinguish between different kinds of leaders. There are leaders that gain their persuasiveness through appeals to morality. Think of someone like Gandhi trying to get people to reach a higher morality and a higher understanding of relations—in this case, between groups that had battled each other for many, many years. It seems to be that there are also leaders who try to absorb uncertainty by creating a vision of the world and persuading their followers to see reality the way they see it. There is potential for innovation there, but there is also potential for great harm. There are many leaders who create a vision of the world in order to reinforce their position no matter what the other consequences may be; and, of course, some of these consequences may be very destructive.

JAMES MARCH: Let me say just a few words about integrity and identity. One of the problems with the Don Quixote message in modern life is that we tend to

have a notion that when I say, "I know who I am," it is about individual, unsocial persons who have created an identity for themselves. And that's not the Quixote vision at all. The vision is one of an identity that is deeply social. Don Quixote is always negotiating with the social meaning of being a knight errant. He is not able to say what a knight errant does arbitrarily, he's trying to figure out what is the meaning of being a knight errant? And the meaning of being a knight errant is not his to choose. If you put it in modern terms and say, "I am a teacher, I should do what teachers do," you cannot arbitrarily choose what teachers do, you have to go to what the meaning of that tradition is, to try to figure out what is the meaning and the essence of being a teacher? That's what Quixote is doing. What is the essence of being a knight errant? It is *not* something that one can just decide; it is something that is deeply embedded socially.

Audience Member: It was mentioned that the business world was more mundane that some of these other pursuits. Many of use are in the business world, so I can certainly agree with you. And that's the world where we are trying to find some meaning and trying to play out some of these themes. I remember back in the late 1970s and early 1980s when I worked in Citibank. And that was a very exciting time for me—I found that it was joyful in many ways. It was exciting, it was innovating, it felt like it was an incredible fun place to work. I know you [John Reed] were there then. I would like to ask about the extent to which, in your leadership role at Citi in those days, these themes play out, particular around joy and imagination?

John Reed: We were lucky at that time, we were building businesses. Nowadays in America they are consolidating businesses and rationalizing businesses and there's been almost no revenue growth that doesn't come from accounting. [laughs] I mean, really, if you look at the last quarterly reports, an awful lot is releasing of reserves of this and that and the other. We were the joy and the vision, we had a vision different from banking as it had been, particularly on the consumer side. We were trying to build something new but also globally and internationally. And we had a lot of fun. The compensation for us was the people with whom we worked, and that which we were trying to create, and it wasn't the times of stock options and bonuses and great monetary rewards that somehow became the goal of many business students later on. The world that Jim refers to is very much that world and I consider myself very lucky to have been able to participate and in some ways been a contributor to that because it was a lot of fun—a lot more fun than most people are having today in the business world. And it is unfortunate that that just happens to be the way it is. There is no question that we had a vision, it was different, and we were trying to build it. We had lots of failures, lots of write-offs—but we had permission to experiment and to bring something new. And certainly, I think the biggest reward was the joy of working with each other and the experience of doing that. I'm sort of sorry for this generation of management that doesn't seem to be operating in that kind of mood, because it really was quite different. And from a leadership point of view, you would much rather lead an enterprise that is driven by creativity and growth and bringing new people on from differ-

ent disciplines and trying to find new ways of doing things—as opposed to this world of consolidation and rationalization in which we are currently operating.

So, very different worlds—they call for very different kinds of leadership. I don't know if the leaders have changed. Look at Sony. Sony in the 1970s and 1980s was one of these great creative companies. Now they are rationalizing their business and trying to figure out how to compete in a very difficult world than they were operating in. And is there room for a visionary leader? I believe so. Do we see one or is that part of the world we are operating in? Probably not, although Silicon Valley has always kept a certain amount of that.

Audience Member: This question is for each of the panelists. Who, if you had to choose one individual, would be the person you most respected as a leader?

David Teece: Well, Winston Churchill is one of my favorites. And why Winston Churchill? Because of his commitment and determination to defeat Nazi Germany, and his ability to communicate that commitment, his ability to give hope in the face of adversity, his utter commitment to success whatever the cost. Also, look at the pictures of him during the war years. There's a lot of joy there, despite the fact that he was working against these overwhelming odds. There was his clear confidence that Nazi Germany would eventually be defeated. To me, Churchill had great leadership skills.

Eric Wanner: My personal favorite would be Mahatma Gandhi—not because he aspired to a higher set of moral ideals, but because he really developed a new way of approaching political conflict by trying to resolve it nonviolently. That had all kinds of implications, running through the whole civil rights movement most prominently, but also in many other areas. And since we live in a world so populated by conflict, the idea of having some nonviolent way of resolving conflict seems to me as a very important thing, a very important vision. And he started that vision—a vision that may have long-run consequences for centuries.

John Reed: I wouldn't be so brave as to have a considered opinion on "a favorite leader." Certainly Gandhi, for reasons that Eric mentioned, you'd have to put on the list. Somebody who I toss out not just in a controversial sense but who I've always admired was Mao in China. Here was a person who was able, after the Japanese were thrown out, and after the Westerners were thrown out by him, to consolidate a country that historically has not consolidated. And with great frustration and errors. One thing that I've always appreciated about Mao is that he made some horrible mistakes. I can just sense the frustration that led to the Cultural Revolution, the inability to really move the society the way he wanted it to go. So he did something that turned out to be inherently evil—he just sort of blew the place apart with the confidence that maybe it could get rebuilt in a better way. So the fact that he failed in some ways, but ultimately succeeded because the China that we see today is the only country in the Asian Pacific that has in fact been able to manage succession (post-World War II) through a set of individuals. And they have, unlike Singapore or Indonesia, been able (in a post-Mao way) not to get stuck. And while there's much evil and bad that came from Mao, I think he has done a lot for a country that had gone

through 500 lousy years and is on the way to modernization. He failed and was frustrated, but nonetheless somehow planted the seeds for success. In spite of what we can ascribe to him of negative things, I think on balance he really was an interesting leader.

JAMES MARCH: I don't have any quarrels with any of this except that all of those are people that I only know by the mythology around them. So it is harder for me because many of the things that I value tend to get lost in the mythology. There are actually a couple of people I would put on the list. Jerry Beasley, a fellow who got educated at Harvard and Stanford and went back to West Virginia to try to bring higher education to kids from the country of West Virginia. And Cory Booker who was at Stanford and Oxford and Yale and went back to Newark and tried to do better for people there. Both of them are facing enormous problems and enormous failures and are sticking to it. They aren't as famous and probably won't ever be as famous, but for me they are pretty important models.[42]

Notes

1. N. Machiavelli, *The Prince*, translated by W.K. Marriott, Chapter 3.
2. This is Barnard's emphasis. C. Barnard, *The Function of the Executive* (Cambridge, MA: Harvard University Press, 1938). This view was later elaborated by March and Simon and by Simon. J.G. March and H. Simon: *Organizations* (Oxford: Blackwell, 1958); H. Simon, *Models of My Life* (Cambridge, MA: MIT Press, 1991).
3. C. Barnard, *Dilemmas of Leadership in the Democratic Process* (Princeton, NJ: Princeton University Press, 1948), p. 80.
4. Ralph Stogdill, *Handbook of Leadership* (New York, NY: The Free Press, 1974), p. vii. Other critics of leadership research include the psychologist Jerry Hunt, who noted: "What is missing, in addition to quantity of theoretical formulations or models is a "grand" or generalized theory of leader-subordinate relationship—if indeed, such a theoretical development is possible." J. Hunt and L. Larson, *Leadership: The Cutting Edge* (Carbondale, IL: Southern Illinois University Press, 1977).
5. Joseph Rost, *Leadership for the Twenty-First Century* (Westport, CT: Praeger, 1991).
6. Ibid.
7. P. Drucker, *The Principles of Management* (New York, NY: HarperCollins Publishers, 1954), p. 158.
8. P. Selznick, *Leadership in Administration* (New York, NY: Harper & Row Publishers, 1957).
9. B. Hermalin, "Toward an Economic Theory of Leadership: Leading by Example," *American Economic Review*, 88 (1998): 1188-1206.
10. M. Casson, *Economics of Business Culture: Game Theory, Transaction Costs and Economic Performance* (Oxford: Clarendon Press, 1991).
11. J. Gardner, *On Leadership* (New York, NY: Free Press, 1990), p. 1.
12. The dynamic capability perspective was developed in D. Teece, G. Pisano, and A. Shuen, "Dynamic Capabilities and Strategic Management," *Strategic Management Journal*, 18/7 (1997): 537-533. Also see S. Winter, "Understanding Dynamic Capabilities," *Strategic Management Journal*, 24/10 (2003): 991-995; D. Teece, "Explicating Dynamic Capabilities," working paper, 2003.
13. J. Schumpeter, *History of Economic Analysis* (New York, NY: Oxford University Press, 1954), p. 41.
14. J. Schumpeter, *The Theory of Economic Development* (Cambridge: Harvard University Press, 1934), p. 91. This is the essence of Schumpeter's entrepreneurship; in full length quoted as follows: "First of all, there is the dream and the will to found a private kingdom, usually, though not necessarily, also a dynasty. Then there is the will to conquer: the impulse to fight, to prove oneself superior to others, to succeed for the sake, not of the fruits of success, but of success itself. From this aspect, economic action becomes akin to sport. Finally, there

is the joy of creating, of getting things done, or simply of exercising one's energy and ingenuity." It is worth noting that the entrepreneurial/leadership function is creative—not "equilibrating" as in other conceptions of entrepreneurship (in particular Israel Kirzner's). The reason is that entrepreneurship is an evolutionary process of creative destruction; one that "incessantly revolutionizes the economic system from within, incessantly destroying the old one, incessantly creating a new one. This process of Creative Destruction is the essential fact about capitalism." (p. 83.)

15. M. Augier, J. March, and B. Sullivan, "Notes on the Evolution of a Research Community: Organization Studies in Anglophone North America 1945-2000," *Organization Science* (forthcoming 2005).
16. D. Teece, "Economic Analysis and Strategic Management," *California Management Review*, 26/3 (1984): 87-110.
17. As David Kreps recently noted: "I am increasingly convinced that economists should—and will—have to change large pieces of the paradigm that has kept us relatively monolithic for the past fifty years. We'll increasingly look like and work with our colleagues in the other . . . social sciences." D. Kreps, "Beliefs and Tastes: Confessions of an Economist," in M. Augier and J.G. March, eds., *Models of a Man: Essays in Memory of Herbert Simon* (Cambridge, MA: MIT Press, 2004). Indeed, understanding the fundamental issues of management requires much more than just economics, sociology, or psychology.
18. J.G. March, "Exploration and Exploitation in Organizational Learning," *Organization Science*, 2 (1991): 71-87.
19. See, for instance, A. Huff, "Citigroup's John Reed and Stanford's James March on Management Research and Practice," *Academy of Management Executive*, 14/1 (2000): 52-64; Teece (2003), op. cit.
20. J.G. March, "A Scholars Quest," *Stanford Business School Magazine*, 64/4 (1996): 10-13.
21. M. Augier, "Making Management Matter: An Interview with John S. Reed," *Academy of Management Learning and Education* (forthcoming 2005), p. 10.
22. J. March and R. Sutton, "Organizational Performance as a Dependent Variable," *Organization Science*, 8 (1997): 697-706.
23. This section builds on Augier, March, and Sullivan (2005), op. cit., and Teece (1984), op. cit.
24. Simon, op. cit., p. 138.
25. S. Leslie, *The Cold War and American Science* (New York, NY: Columbia University Press, 1993); see Augier, March, and Sullivan (2005), op. cit.
26. For a longer discussion of these issues and the implications for the field of organization theory, see Augier, March, and Sullivan (2005), op. cit; for the report, see R.A. Gordon and J.E. Howell, *Higher Education for Business* (New York, NY: Columbia University Press 1959).
27. S. Schlossman, M. Sedlak, and H. Wechsler, "The "New Look": The Ford Foundation and the Revolution in Business Education," *Selections*, 14/3 (1987): 8-28.
28. The Ford Foundation Behavioral Science Program, "Proposed Plan for the Development of the Behavioral Sciences Program," Herbert A. Simon papers, Carnegie Mellon University Library, 1951.
29. Ibid., p. 4.
30. March and Simon, op. cit.; R. Cyert and J.G. March, *A Behavioral Theory of the Firm*, 2nd edition (Oxford: Blackwell, 1992).
31. The movie, "Passion and Discipline," can be purchased through "Films for the Humanities and Sciences," www.films.com. A flavor of some of the main ideas in the movie and the background is also discussed in James G. March and Mie Augier, "James March on Education, Leadership, and Don Quixote: Introduction and Interview," *Academy of Management Learning & Education*, 3/2 (June 2004): 169-177.
32. See, in particular, March (1991), op. cit.; J.G. March, "Continuity and Change in Theories of Organizational Action," *Administrative Science Quarterly*, 41 (1996): 278-287.
33. For an application of this idea to the study of research programs, see March [(1991), op. cit.]: "[Research programs and ideas, such as those prevailing business education] that engage in exploration to the exclusion of exploitation are likely to find that they suffer the cost of experimentation without gaining many of its benefits. They exhibit too many undeveloped new ideas and too little distinctive competence. Conversely, [research programs and ideas] that engage in exploitation to the exclusion of exploration are likely to find themselves trapped in suboptimal stable equilibria." [p. 71.]
34. March (1996), op. cit.
35. March (1991), op. cit., March (1996), op. cit.; Teece (2003), op. cit.

36. March (1999), op. cit.
37. A.N. Whitehead, *Modes of Thought* (New York, NY: Macmillan, 1938), p. 15.
38. "The [leader] does not deduce his authority from codes and statutes, as is the case with the jurisdiction of office; nor does he deduce his authority from traditional custom or feudal vows of faith, as in the case with patrimonial power."
39. Schumpeter (1949), op. cit., pp. 71-72.
40. Imagination, Commitment, and Joy are the three main themes/sections in Professor March's film on Leadership.
41. J. Brown, *Seeing Things Differently: Perspectives on Innovation* (Boston, MA: Harvard University Press, 1998).
42. Jerry Beasley is the President of Concord College, and Cory Booker is a Newark City Councilman. They were both part of Professor March's film.